

Pavan Teja Tallapalli

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Summary

Software engineer with experience in backend development, SDK integration, API engineering, application testing, and automation across telecom platforms. At Jio Platforms, contributed to Heartbeat, Data Collector, and JIOoSCC initiatives by improving device-side software reliability, validation workflows, and production diagnostics. Additional experience at BSNL in MQTT v5 API development, eSIM LPA SDK integration, protocol-level troubleshooting, and security-focused testing. Currently pursuing an M.S. in Data Science at Arizona State University. Delivered measurable impact through support for 8,000+ connected devices, a 40% reduction in false alerts, and analytics on 100K+ transaction records.

Technical Skills

- **Languages:** Python, C, C++, Go, Java, JavaScript, C#
- **Development:** REST APIs, Object-Oriented Programming, data structures and algorithms, Flask, Node.js, ReactJS, HTML5, DevOps, CI/CD, software development life cycle, Angular, full-stack, frontend, backend, GraphQL, SCRUM
- **Testing & QA:** Selenium, Postman, Swagger, API Testing, Unit Testing, Integration Testing, Regression Testing, microservices, debugging, A/B testing, Valgrind, Jira, Software Engineering Practices, Zapier, Make, Workato
- **Data & Databases:** MySQL, PostgreSQL, MongoDB, Pandas, Golang, Snowflake, PySpark, PyTorch, TensorFlow, Scala
- **Cloud & Tools:** AWS S3, GitHub, Linux, Jenkins, Docker, Azure, GCP, GCS, Terraform, Kubernetes
- **Other:** Agile, LLM APIs, Spring Boot, OAuth2, Rust, Django, Kafka, FastAPI, Big Data, DuckDB, Galileo, ADLC, Microsoft Dynamics 365 CRM

Professional Experience

Software Development Engineer, Jio Platforms Limited, India

June 2023 – July 2024

- Designed and optimized **SDK modules, WebUI components, and Salesforce-integrated workflows**, collaborating with cross-functional teams to improve system performance, usability, and platform reliability.
- Contributed to development and testing across 3 production-facing telecom components, strengthening observability, validating functional behavior, and improving release quality in production environments to meet customer needs.
- Automated RSA key generation and encryption workflows and implemented an **automation testing framework for SDK and Salesforce-integrated workflows**, improving testing coverage and release efficiency.
- Diagnosed defects using telemetry, logs, and monitoring data from multiple production workflows, accelerating issue reproduction, root-cause analysis, and validation of fixes during development and QA cycles.
- Collaborated with vendor, product, and engineering teams across device, SDK, and platform integration tracks to resolve issues based on user feedback and support stable software releases for Jio AirFiber deployments.
- Supported deployment of 1 TensorFlow Lite anomaly-detection model on edge devices that reduced false alerts by 40% and improved proactive monitoring of abnormal operational behavior.

Software Engineer (Networking Systems), Bharat Sanchar Nigam Limited, India

June 2022 – June 2023

- Implemented MQTT v5 messaging workflows and REST-based APIs for eSIM LPA SDK services supporting 4 lifecycle operations, download, activation, deactivation, and deletion across 8,000+ connected devices.
- Engineered C/C++ modules for GSM/LTE switching systems, improving efficiency across telecom software operations.
- Designed and executed unit, integration, API, and regression tests for networking modules and SDK workflows, improving defect isolation and increasing deployment confidence across multiple release cycles.
- Performed packet inspection, protocol analysis, and security validation with 5 tools, Wireshark, TShark, TCPdump, Postman, and Burp Suite to isolate interoperability and request-response issues.
- Collaborated with development and QA teams to document reproducible defects, validate fixes, and support releases.

Projects

Crispr scrollytelling

September 2025 – December 2025

- Developed interactive data visualizations using D3.js and Scrollama to analyze 3 major dimensions of CRISPR, research, regulation, and adoption through narrative visual storytelling.

Ai-powered personal finance agent

April 2024 – August 2024

- Engineered a secure financial microservice (FastAPI, React, SQLite) and automated ETL pipeline, utilizing SMOTE resampling, and deployed a Gradient Boosting fraud classifier achieving a 0.93 F1-score and 0.98 ROC-AUC.
- Orchestrated a conversational GenAI copilot (Llama 3.2, LangGraph, BART) utilizing the Model Context Protocol (MCP) and Vector Databases for real-time inference, coupled with an unsupervised anomaly pipeline (Isolation Forest, Z-score) to deliver 0.926 NLP intent accuracy and a 218ms median query latency.

Education

Master of Science, Arizona State University, USA

August 2024 – May 2026

Data Science (Computing and Decision Analytics)

Bachelor of Technology, Jaypee Institute of Information Technology, India

July 2019 – May 2023

Electronics and Communications Engineering (AI/ML)